Phase 0 of the simplex algorithm

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The simplex auxiliary problem

Introduction: The simplex auxiliary problem

- The simplex algorithm starts from a **basic feasible solution**: when the slack variables do not give one, the algorithm cannot be started
- The **Phase** 0 is the procedure that finds a first basic feasible solution, or proves that there is none
- It is based on a new LP problem, the **simplex auxiliary problem**, obtained by adding one variable x_0
- We define it, and we see that it is always solvable and that its optimal value tells whether the original LP problem is feasible

Standard maximization form

- We consider an LP problem in standard maximization form with $n + m$ non-negative variables and m equality linear constraints:

$$\begin{aligned} \max \quad & \boldsymbol{c}' \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0. \end{aligned}$$

- It is obtained from an LP problem in canonical maximization form with n non-negative variables and m less-than-or-equal-to linear constraints by adding m slack non-negative variables

The initial basic feasible solution

- The Step 0 of the simplex algorithm requires an initial basic feasible solution
- If $\boldsymbol{b} \geq 0$, the partition

$$\mathcal{B} = \{n+1, n+2, \ldots, n+m\} \quad \text{and} \quad \mathcal{N} = \{1, 2, \ldots, n\}$$

  is associated with the basis $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and with the initial basic feasible solution $\boldsymbol{x}_{\mathcal{B}} = \boldsymbol{b}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$
- If instead $b_i < 0$ for some $i \in \{1, 2, \ldots, m\}$, this partition is not associated with a basic feasible solution, and the simplex algorithm cannot be started
- In this case we execute the **Phase** 0 of the simplex algorithm

Phase 0 of the simplex algorithm

Definition 1: Phase 0 of the simplex algorithm

*The **Phase** 0 of the simplex algorithm is a procedure which determines a basic feasible solution of the LP problem in standard maximization form, together with the associated dictionary, or proves that the LP problem is **infeasible**.*

To do so, we build a new LP problem
with one more variable, which always has
a basic feasible solution that is easy to find.

Simplex auxiliary problem

Definition 2: simplex auxiliary problem

Given an LP problem in standard maximization form, the associated ***simplex auxiliary problem*** *is the LP problem in standard maximization form obtained by adding one non-negative variable x_0, called the* ***auxiliary variable****, subtracting it from the left-hand side of every linear constraint, and replacing the objective function:*

$$\max \quad -x_0$$

$$\textit{subject to} \quad \sum_{j=1}^{n+m} a_{ij}\, x_j - x_0 = b_i, \qquad \forall i \in \{1, 2, \ldots, m\}$$

$$x_0,\ x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n+m\}$$

Input data of the simplex auxiliary problem (1/3)

- The auxiliary variable x_0 is placed in the first position
- The input data of the simplex auxiliary problem are:
  - the column vector $\boldsymbol{x} \in \mathbb{R}^{n+m+1}$ of the variables
  - the objective-function vector $\boldsymbol{c} \in \mathbb{Z}^{n+m+1}$, whose first entry is -1 and whose other entries are 0
  - the constraint matrix $\boldsymbol{A} \in \mathbb{Z}^{m \times (n+m+1)}$, obtained by prepending to the original constraint matrix a column of m entries all equal to -1
  - the right-hand-side vector $\boldsymbol{b} \in \mathbb{Z}^m$

Input data of the simplex auxiliary problem (2/3)

$$\boldsymbol{x} = \begin{pmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \\ x_{n+1} \\ \vdots \\ x_{n+m} \end{pmatrix} \qquad \boldsymbol{c} = \begin{pmatrix} -1 \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{pmatrix}$$

- The first entry of $\boldsymbol{x}$ is the auxiliary variable x_0, and the objective function is $\boldsymbol{c}' \boldsymbol{x} = -x_0$

Input data of the simplex auxiliary problem (3/3)

$$\boldsymbol{A} = \begin{pmatrix} -1 & a_{11} & \dots & a_{1n} & 1 & \dots & 0 \\ -1 & a_{21} & \dots & a_{2n} & 0 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots & \vdots & \dots & \vdots \\ -1 & a_{m1} & \dots & a_{mn} & 0 & \dots & 1 \end{pmatrix} \qquad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

- The first column of $\boldsymbol{A}$ is the column of x_0, the last m columns are the identity matrix of the slack variables

The simplex auxiliary problem is always solvable

The simplex auxiliary problem is never infeasible and it is never unbounded:

1. it always admits a feasible solution, obtained by setting $x_j = 0$ for every $j \in \{1, 2, \ldots, n\}$,

$$x_0 = \max\{0, -\min_i b_i\}$$

   and the slack variables $x_{n+i} = b_i + x_0 \geq 0$ for every $i \in \{1, 2, \ldots, m\}$

2. its objective value is $-x_0 \leq 0$ for every feasible solution

Accordingly, by the Fundamental Theorem of Linear Programming, it always admits an **optimal solution**.

Feasibility of an LP problem

Theorem 1: feasibility of an LP problem

The LP problem in standard maximization form admits a feasible solution if and only if the optimal value of the associated simplex auxiliary problem is equal to 0.

- The idea of the proof is that a feasible solution of the LP problem is exactly a feasible solution of the simplex auxiliary problem in which the auxiliary variable takes the value $x_0 = 0$

Proof (1/2)

We prove the two implications separately.

($\Rightarrow$) Let $\bar{\boldsymbol{x}}$ be a feasible solution of the LP problem. The point obtained by adding $\bar{x}_0 = 0$ is a feasible solution of the simplex auxiliary problem, since

$$\sum_{j=1}^{n+m} a_{ij}\,\bar{x}_j - \underbrace{\bar{x}_0}_{=0} = b_i, \qquad \forall i \in \{1, 2, \ldots, m\},$$

and its objective value is $-\bar{x}_0 = 0$.

Since the objective value of every feasible solution of the simplex auxiliary problem is $-x_0 \leq 0$, this solution is optimal and the optimal value is 0.

Proof (2/2)

($\Leftarrow$) Let the optimal value of the simplex auxiliary problem be equal to 0 and let $\tilde{\boldsymbol{x}}$ be an optimal solution. Then $\tilde{x}_0 = 0$ and, by the constraints of the simplex auxiliary problem,

$$\sum_{j=1}^{n+m} a_{ij}\, \tilde{x}_j = b_i + \underbrace{\tilde{x}_0}_{=0} = b_i, \qquad \forall i \in \{1, 2, \ldots, m\}.$$

Accordingly, the point obtained by removing the auxiliary variable satisfies the equality constraints $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$ and the non-negativity constraints $\boldsymbol{x} \geq 0$, and then it is a feasible solution of the LP problem. □

A certificate of infeasibility

If the optimal value of the simplex auxiliary problem
is strictly negative,

then the LP problem is **infeasible**.

Let us now build the simplex auxiliary problem
of a small LP problem with $n = 2$ and $m = 3$.

Example (1/3)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrrrrrrr}
\max & & x_1 & + & 2 & x_2 & & \\
\text{subject to} & - & x_1 & - & & x_2 & \leq & -3 \\
& - & x_1 & + & & x_2 & \leq & 2 \\
& & x_1 & + & & x_2 & \leq & 6 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$

- We use this LP problem as the guiding example of the chapter

Example (2/3)

By adding $m = 3$ slack variables, we obtain the following LP problem in standard maximization form with $n + m = 5$ non-negative variables:

$$
\begin{array}{rrrrrrrrrrrrr}
\max & x_1 & + & 2 & x_2 & & & & & & & & \\
\text{subject to} & -x_1 & - & & x_2 & + & x_3 & & & & & = & -3 \\
& -x_1 & + & & x_2 & & & + & x_4 & & & = & 2 \\
& x_1 & + & & x_2 & & & & & + & x_5 & = & 6 \\
& x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

- Since $b_1 = -3 < 0$, the partition $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2\}$ is not associated with a basic feasible solution, and we have to execute the Phase 0

Example (3/3)

The associated simplex auxiliary problem with $n + m + 1 = 6$ non-negative variables reads as follows:

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & - & x_0 & & & & & & & & & & & & \\
\text{subject to} & - & x_0 & - & x_1 & - & x_2 & + & x_3 & & & & & = & -3 \\
& - & x_0 & - & x_1 & + & x_2 & & & + & x_4 & & & = & 2 \\
& - & x_0 & + & x_1 & + & x_2 & & & & & + & x_5 & = & 6 \\
& & x_0, & & x_1, & & x_2, & & x_3, & & x_4, & & x_5 & \geq & 0
\end{array}
$$

- The column of x_0 has all its entries equal to -1 and the objective function is $-x_0$

Summary: the simplex auxiliary problem

When the Phase 0 is needed:

- If $b_i < 0$ for some i, the slack variables do not give a basic feasible solution
- The Phase 0 finds one, with its dictionary, or proves that the LP problem is infeasible

The simplex auxiliary problem:

- Add $x_0 \geq 0$, subtract it from every constraint, and maximize $-x_0$
- It is never infeasible and never unbounded
- The LP problem is feasible if and only if its optimal value is 0

Solving the simplex auxiliary problem

Introduction: Solving the simplex auxiliary problem

- The simplex auxiliary problem is solved with the simplex algorithm itself
- Its first dictionary is not feasible either, but a single **special pivoting operation** on x_0 makes it feasible
- Then the simplex algorithm runs as usual until an optimal dictionary is found
- We see the three possible outcomes of the Phase 0, and we execute it on the guiding example

The initial dictionary of the simplex auxiliary problem

- The m slack variables give the partition

$$\mathcal{B} = \{n+1, n+2, \ldots, n+m\} \quad \text{and} \quad \mathcal{N} = \{0, 1, 2, \ldots, n\},$$

which is associated with the basis $\boldsymbol{A}_{\mathcal{B}} = \boldsymbol{I}$ and with the dictionary

$$\begin{cases} x_i = b_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij}\, x_j, \quad \forall i \in \mathcal{B} \\ \\ -w = 0 - x_0 \end{cases}$$

- $-w = -x_0$ denotes the objective function of the simplex auxiliary problem (i.e., $w = x_0$)
- This dictionary is not associated with a basic feasible solution, since $b_i < 0$ for some $i \in \mathcal{B}$

- A single **special pivoting operation** is enough to obtain a basic feasible solution of the simplex auxiliary problem
- The auxiliary variable x_0 **enters** the basis
- The basic variable **leaving** the basis is the one with the most negative value

Special pivoting operation

- The pivot column is the auxiliary variable x_0 and the pivot row is the basic variable x_r with

$$r \in \arg\min\{\, b_i : i \in \mathcal{B} \,\}$$

- After this pivoting operation, the values of the variables are

$$x_0 = -b_r \qquad \text{and} \qquad x_i = b_i - b_r, \quad \forall i \in \mathcal{B} \setminus \{r\}$$

- They are all non-negative, since $b_r \leq b_i$ for every $i \in \mathcal{B}$

Then the simplex algorithm

- Starting from the basic feasible solution obtained with the special pivoting operation, the simplex algorithm is executed on the simplex auxiliary problem until an optimal solution is found
- Since the simplex auxiliary problem is never unbounded, the algorithm always terminates with an **optimal dictionary**

What does the optimal dictionary tell us
about the original LP problem?

The three cases at the end of the Phase 0

At the end of the Phase 0, exactly one of the following three cases occurs:

1. the optimal value of the simplex auxiliary problem is **strictly negative**: the LP problem is **infeasible**
2. the optimal value is 0 and the auxiliary variable x_0 is **non-basic**: the optimal basis of the simplex auxiliary problem, after removing x_0 from $\mathcal{N}$, is a basis of the LP problem associated with a basic feasible solution
3. the optimal value is 0 and the auxiliary variable x_0 is **basic**, necessarily with value $x_0 = 0$: one further pivoting operation makes x_0 leave the basis, bringing back to the previous case

The third case: x_0 basic with value 0 (1/2)

- In the third case, the basic feasible solution of the simplex auxiliary problem is **degenerate**, since the basic variable x_0 takes the value 0
- A pivoting operation is performed on the row of x_0, choosing as pivot column any non-basic variable x_j with $j \in \mathcal{N}$ such that

$$\bar{a}_{0j} \neq 0$$

- The value of the other basic variables does not change, since the value of x_0 is 0

The third case: x_0 basic with value 0 (2/2)

- Such a variable x_j always exists
- Otherwise, the row of x_0 in

$$\boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{A}$$

(without the column of x_0) would be zero

- This contradicts $\text{rank}(\boldsymbol{A}) = m$, which holds since $\boldsymbol{A}$ contains the identity matrix of the slack variables

Let us now execute the Phase 0
on the simplex auxiliary problem
of the guiding example.

Example: Phase 0 (1/5)

- Consider the simplex auxiliary problem of the guiding example, with $\mathcal{B} = \{3, \ 4, \ 5\}$ and $\mathcal{N} = \{0, \ 1, \ 2\}$
- The initial dictionary is:

$$\begin{cases} x_3 = -3 + x_0 + x_1 + x_2 \\ x_4 = 2 + x_0 + x_1 - x_2 \\ x_5 = 6 + x_0 - x_1 - x_2 \\ -w = 0 - x_0 \end{cases}$$

- The associated basic solution is not feasible, since $\bar{b}_3 = -3 < 0$

Example: Phase 0 (2/5)

Special pivoting operation

- The pivot column is x_0 and the pivot row is x_3, the basic variable with the most negative value
- By isolating x_0 in the first equation we obtain $x_0 = 3 + x_3 - x_1 - x_2$
- By substituting it in the other equations we obtain the dictionary with $\mathcal{B} = \{0, \ 4, \ 5\}$ and $\mathcal{N} = \{1, \ 2, \ 3\}$:

$$\begin{cases} x_0 = 3 - x_1 - x_2 + x_3 \\ x_4 = 5 - 2x_2 + x_3 \\ x_5 = 9 - 2x_1 - 2x_2 + x_3 \\ -w = -3 + x_1 + x_2 - x_3 \end{cases}$$

Example: Phase 0 (3/5)

- The basic feasible solution associated with the dictionary after the special pivoting operation is

$$\bar{x}_0 = 3, \quad \bar{x}_4 = 5, \quad \bar{x}_5 = 9 \quad \text{and} \quad \bar{x}_1 = \bar{x}_2 = \bar{x}_3 = 0$$

  of value $-w = -3$

- As expected from the special pivoting operation, $\bar{x}_0 = -(-3)$, $\bar{x}_4 = 2 - (-3)$ and $\bar{x}_5 = 6 - (-3)$ are all non-negative

- The reduced costs of x_1 and x_2 are positive: the dictionary is not optimal yet, and we continue with the simplex algorithm

Example: Phase 0 (4/5)

First pivoting operation

- The reduced cost of x_1 is $\bar{c}_1 = 1 > 0$, and then the pivot column is x_1
- The values of its column are $\bar{a}_{01} = 1$ and $\bar{a}_{51} = 2$, and then $t_m = \min\left\{\frac{3}{1}, \frac{9}{2}\right\} = 3$ and the pivot row is x_0
- The new dictionary, with $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{0, \ 2, \ 3\}$, is:

$$\begin{cases} x_1 = 3 - x_0 - x_2 + x_3 \\ x_4 = 5 - 2x_2 + x_3 \\ x_5 = 3 + 2x_0 - x_3 \\ -w = 0 - x_0 \end{cases}$$

Example: Phase 0 (5/5)

- All the reduced costs are non-positive, and then this dictionary is optimal
- The optimal value of the simplex auxiliary problem is 0 and the auxiliary variable x_0 is non-basic
- Accordingly, the LP problem of the guiding example is **feasible** and

$$\mathcal{B} = \{1, \quad 4, \quad 5\}$$

is a basis associated with a basic feasible solution

Summary: solving the simplex auxiliary problem

Special pivoting operation:

- x_0 enters the basis, the basic variable with the most negative value leaves it
- After it, $x_0 = -b_r$ and $x_i = b_i - b_r \geq 0$: the dictionary is feasible

End of the Phase 0:

- Optimal value < 0: the LP problem is infeasible
- Optimal value 0, x_0 non-basic: a basic feasible solution of the LP problem
- Optimal value 0, x_0 basic (value 0): one more pivot on its row

From the Phase 0 to the simplex algorithm

Introduction: From the Phase 0 to the simplex algorithm

- The Phase 0 ends with a basis of the LP problem, but its dictionary still contains x_0 and the objective function $-w$ of the simplex auxiliary problem
- We see how to build the **initial dictionary** of the original LP problem from it
- Then the simplex algorithm starts as usual: we solve the guiding example up to its optimal solution, and we follow the path on the feasible region

The optimal dictionary of the Phase 0

- At the end of the Phase 0, the optimal dictionary of the simplex auxiliary problem gives the equations of the basic variables of the LP problem in function of its non-basic variables, once the auxiliary variable x_0 is removed from $\mathcal{N}$
- The last equation of that dictionary, however, is the one of the objective function $-w$ of the simplex auxiliary problem
- It has to be replaced by the equation of the original objective function z

The original objective function

- The equation of the original objective function is obtained by **substituting**, in

$$z = \sum_{j=1}^{n+m} c_j \, x_j,$$

the equation of every basic variable x_i with $i \in \mathcal{B}$, so that z is written in function of the non-basic variables only

- Equivalently, it is computed as

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} \qquad \text{and} \qquad \bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}}$$

- Once the initial dictionary of the LP problem is available, the simplex algorithm is executed as usual, starting from its Step 1

Example: from the Phase 0 to the simplex algorithm (1/2)

- Consider the optimal dictionary of the simplex auxiliary problem of the guiding example, with $\mathcal{B} = \{1, \ 4, \ 5\}$
- By removing the auxiliary variable x_0, the equations of the basic variables of the LP problem are:

$$\begin{cases} x_1 = 3 - x_2 + x_3 \\ x_4 = 5 - 2\,x_2 + x_3 \\ x_5 = 3 - x_3 \end{cases}$$

- The objective function is $z = x_1 + 2\,x_2$, and the basic variable x_1 appears in it
- By substituting its equation, we obtain:

$$z = \underbrace{3 - x_2 + x_3}_{= x_1} + 2\,x_2 = 3 + x_2 + x_3$$

Example: from the Phase 0 to the simplex algorithm (2/2)

- Accordingly, with $\mathcal{B} = \{1, \ 4, \ 5\}$ and $\mathcal{N} = \{2, \ 3\}$, the initial dictionary of the LP problem of the guiding example is:

$$\begin{cases} x_1 = 3 - x_2 + x_3 \\ x_4 = 5 - 2x_2 + x_3 \\ x_5 = 3 - x_3 \\ z = 3 + x_2 + x_3 \end{cases}$$

- The associated basic feasible solution is

$$\bar{x}_1 = 3, \quad \bar{x}_4 = 5, \quad \bar{x}_5 = 3 \quad \text{and} \quad \bar{x}_2 = \bar{x}_3 = 0$$

of value 3

Starting from this initial dictionary,
determined by the Phase 0,
we now execute the simplex algorithm.

Example: the simplex algorithm after the Phase 0 (1/4)

First pivoting operation

- The reduced cost of x_2 is $\bar{c}_2 = 1 > 0$, and then the pivot column is x_2
- The values of its column are $\bar{a}_{12} = 1$ and $\bar{a}_{42} = 2$, and then $t_m = \min\left\{\frac{3}{1}, \frac{5}{2}\right\} = \frac{5}{2}$ and the pivot row is x_4
- The new dictionary, with $\mathcal{B} = \{1, \ 2, \ 5\}$ and $\mathcal{N} = \{3, \ 4\}$, is:

$$\begin{cases} x_1 = \frac{1}{2} + \frac{1}{2}x_3 + \frac{1}{2}x_4 \\ x_2 = \frac{5}{2} + \frac{1}{2}x_3 - \frac{1}{2}x_4 \\ x_5 = 3 - x_3 \\ z = \frac{11}{2} + \frac{3}{2}x_3 - \frac{1}{2}x_4 \end{cases}$$

Example: the simplex algorithm after the Phase 0 (2/4)

Second pivoting operation

- The reduced cost of x_3 is $\bar{c}_3 = \frac{3}{2} > 0$, and then the pivot column is x_3
- The only positive value in its column is $\bar{a}_{53} = 1$, and then $t_m = \frac{3}{1} = 3$ and the pivot row is x_5
- The new dictionary, with $\mathcal{B} = \{1,\ 2,\ 3\}$ and $\mathcal{N} = \{4,\ 5\}$, is:

$$\begin{cases} x_1 = 2 + \frac{1}{2}x_4 - \frac{1}{2}x_5 \\ x_2 = 4 - \frac{1}{2}x_4 - \frac{1}{2}x_5 \\ x_3 = 3 - x_5 \\ z = 10 - \frac{1}{2}x_4 - \frac{3}{2}x_5 \end{cases}$$

Example: the simplex algorithm after the Phase 0 (3/4)

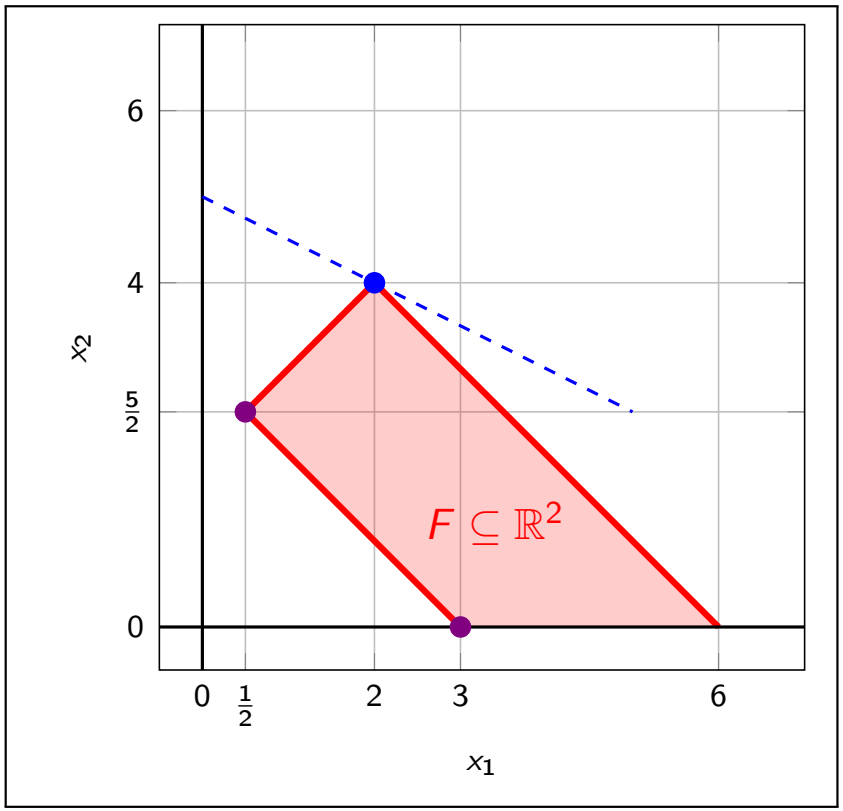
- All the reduced costs are non-positive, and then this dictionary is optimal
- An optimal solution of the LP problem of the guiding example is

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

- The optimal value is

$$z(\,LP\,) = 10$$

Example: the simplex algorithm after the Phase 0 (4/4)



- F is bounded by the lines $-x_1 - x_2 = -3$ (bottom left), $-x_1 + x_2 = 2$ (top left), $x_1 + x_2 = 6$ (right) and the x_1-axis
- Dashed blue: the level set $x_1 + 2\,x_2 = 10$ through the optimal vertex
- Violet: the basic feasible solution $(3, 0)$ of the Phase 0 and the vertex $(\frac{1}{2}, \frac{5}{2})$ after the first pivoting operation
- Blue: the optimal solution

Summary: from the Phase 0 to the simplex algorithm

Initial dictionary of the LP problem:

- Remove x_0 from the optimal dictionary of the Phase 0
- Replace the equation of $-w$ with the one of z, written in function of the non-basic variables (substitution, or $\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{b}$ and $\bar{\boldsymbol{c}}'$)

Then the simplex algorithm:

- It starts from its Step 1
- In the guiding example: from $(3, 0)$, through $(\frac{1}{2}, \frac{5}{2})$, to the optimal solution $(2, 4)$ of value 10

Detecting infeasible LP problems

Introduction: Detecting infeasible LP problems

- The Phase 0 does not only find a starting point: it also **proves** that an LP problem has no feasible solution at all
- This happens when the optimal value of the simplex auxiliary problem is strictly negative
- We see it on a small LP problem with two constraints that contradict each other

Certificate of infeasibility

- If the optimal value of the simplex auxiliary problem is strictly negative, then, by the theorem on the feasibility of an LP problem, the LP problem is **infeasible**
- In this case the optimal dictionary of the simplex auxiliary problem is a **certificate of infeasibility**

Example: infeasible LP problem (1/4)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrcrcr}
\max & x_1 & + & x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 2 \\
- & x_1 & - & x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

- Since $b_2 = -4 < 0$, we execute the Phase 0

Example: infeasible LP problem (2/4)

- With $\mathcal{B} = \{3, \ 4\}$ and $\mathcal{N} = \{0, \ 1, \ 2\}$, the initial dictionary of the simplex auxiliary problem is:

$$\begin{cases} x_3 = 2 + x_0 - x_1 - x_2 \\ x_4 = -4 + x_0 + x_1 + x_2 \\ -w = 0 - x_0 \end{cases}$$

- The associated basic solution is not feasible, since $\bar{b}_4 = -4 < 0$

Example: infeasible LP problem (3/4)

Special pivoting operation

- The pivot column is x_0 and the pivot row is x_4, the basic variable with the most negative value
- The new dictionary, with $\mathcal{B} = \{0, \ 3\}$ and $\mathcal{N} = \{1, \ 2, \ 4\}$, is:

$$\begin{cases} x_0 = 4 - x_1 - x_2 + x_4 \\ x_3 = 6 - 2x_1 - 2x_2 + x_4 \\ -w = -4 + x_1 + x_2 - x_4 \end{cases}$$

Example: infeasible LP problem (4/4)

First pivoting operation

- The pivot column is x_1, since $\bar{c}_1 = 1 > 0$
- The values of its column are $\bar{a}_{01} = 1$ and $\bar{a}_{31} = 2$, and then $t_m = \min\left\{\frac{4}{1}, \frac{6}{2}\right\} = 3$ and the pivot row is x_3
- The new dictionary, with $\mathcal{B} = \{0, \ 1\}$ and $\mathcal{N} = \{2, \ 3, \ 4\}$, is:

$$\begin{cases} x_0 = 1 + \frac{1}{2}x_3 + \frac{1}{2}x_4 \\ x_1 = 3 - x_2 - \frac{1}{2}x_3 + \frac{1}{2}x_4 \\ -w = -1 - \frac{1}{2}x_3 - \frac{1}{2}x_4 \end{cases}$$

Example: infeasible LP problem, conclusion

- All the reduced costs are non-positive, and then this dictionary is optimal
- The optimal value of the simplex auxiliary problem is

$$-1 < 0$$

  that is, the auxiliary variable takes the value $\tilde{x}_0 = 1$ in every optimal solution
- Accordingly, the LP problem of this example is **infeasible**

Summary: detecting infeasible LP problems

Infeasibility:

- Optimal value of the simplex auxiliary problem strictly negative $\Rightarrow$ the LP problem is infeasible
- The optimal dictionary of the simplex auxiliary problem is a certificate of infeasibility

In the example:

- The constraints $x_1 + x_2 \leq 2$ and $-x_1 - x_2 \leq -4$ cannot hold together
- The Phase 0 ends with optimal value -1: the auxiliary variable cannot reach 0

Pseudo-code of the simplex algorithm with the Phase 0

Introduction: Pseudo-code of the simplex algorithm with the Phase 0

- We now put everything together in a single algorithm
- The Phase 0 becomes part of the Step 0 (initialization) of the simplex algorithm: it is executed only when some b_i is negative
- Steps 1 and 2 are the usual optimality check and pivoting operations, used both on the simplex auxiliary problem and on the original LP problem

Pseudo-code (1/5)

Input
An LP problem in standard maximization form.

Output
An optimal solution of the LP problem, if one exists; otherwise, a certificate that the LP problem is infeasible or unbounded.

Step 0: **Initialization**
If $b_i \geq 0$ for all $i \in \{1, 2, \ldots, m\}$, then set

$$\mathcal{B} \leftarrow \{n+1, n+2, \ldots, n+m\},$$
$$\mathcal{N} \leftarrow \{1, 2, \ldots, n\},$$

together with the associated dictionary, and go to Step 1.

Pseudo-code (2/5)

Step 0: **Initialization** (continued)
Otherwise, execute the **Phase** 0:

a. build the simplex auxiliary problem and its initial dictionary with $\mathcal{B} \leftarrow \{n+1, \ldots, n+m\}$ and $\mathcal{N} \leftarrow \{0, 1, \ldots, n\}$;
b. perform the **special pivoting operation**: the pivot column is x_0 and the pivot row is x_r with $r \in \arg\min\{ b_i : i \in \mathcal{B} \}$;
c. execute Step 1 and Step 2 on the simplex auxiliary problem until an optimal dictionary is found;

Pseudo-code (3/5)

Step 0: **Initialization** (continued)

d. if the optimal value is strictly negative, then the LP problem is **infeasible** and `STOP`;
e. if the auxiliary variable x_0 is basic, perform one pivoting operation on its row to make it leave the basis;
f. remove x_0 from $\mathcal{N}$ and replace the equation of $-w$ with the equation of the original objective function z, written in function of the non-basic variables.

Go to Step 1.

Pseudo-code (4/5)

Step 1: **Optimality check**
With respect to the current dictionary, if $\bar{c}_j \leq 0$ for all $j \in \mathcal{N}$, then the basic feasible solution $x_i = \bar{b}_i$, $\forall i \in \mathcal{B}$, and $x_j = 0$, $\forall j \in \mathcal{N}$, is an optimal solution and $\bar{z}$ is the optimal value. In this case `STOP`.

Pseudo-code (5/5)

Step 2: **Pivoting operations**

The **pivot column** is a non-basic variable x_p with $p \in \mathcal{N}$ such that $\bar{c}_p > 0$. If $\bar{a}_{ip} \leq 0$ for all $i \in \mathcal{B}$, then the LP problem is **unbounded** and `STOP`. Otherwise compute

$$t_m \leftarrow \min \left\{ \frac{\bar{b}_i}{\bar{a}_{ip}} : i \in \mathcal{B},\ \bar{a}_{ip} > 0 \right\},$$

determine the **pivot row** x_r with $\bar{a}_{rp} > 0$ and $\bar{b}_r - t_m\, \bar{a}_{rp} = 0$, update the dictionary and set $\mathcal{B} \leftarrow \big(\mathcal{B} \setminus \{r\}\big) \cup \{p\}$ and $\mathcal{N} \leftarrow \big(\mathcal{N} \setminus \{p\}\big) \cup \{r\}$. Go to Step 1.

Summary: pseudo-code

Step 0:

- If $\boldsymbol{b} \geq 0$: start from the slack variables
- Otherwise: Phase 0 (auxiliary problem, special pivot, simplex until optimal, check the optimal value, remove x_0, restore z)

Steps 1 and 2:

- Optimality check: $\bar{c}_j \leq 0$ for all $j \in \mathcal{N}$
- Pivoting: pivot column with $\bar{c}_p > 0$, unboundedness test, ratio test t_m, pivot row

The Fundamental Theorem of Linear Programming

Introduction: The Fundamental Theorem of Linear Programming

- With the Phase 0, the simplex algorithm can be applied to **every** LP problem in standard maximization form
- The Phase 0 and the simplex algorithm, executed one after the other, prove the **Fundamental Theorem of Linear Programming**
- It describes all the possible outcomes of an LP problem and shows that, whenever an optimal solution exists, one of them is a basic feasible solution

Fundamental Theorem of Linear Programming

Theorem 2: Fundamental Theorem of Linear Programming

For every LP problem in standard maximization form:

1. *if it admits a feasible solution, then it admits a* ***basic*** *feasible solution;*
2. *if it admits an optimal solution, then it admits a* ***basic*** *feasible solution which is optimal;*
3. *it is* ***infeasible****, it is* ***unbounded****, or it admits an* ***optimal solution****.*

The idea of the proof

- The Phase 0 and the simplex algorithm, executed with an **anti-cycling** pivot rule, terminate after a finite number of pivoting operations
- Each of their possible outcomes is one of the three cases of the statement

Proof (1/4)

- We execute the Phase 0 on the LP problem, and then the simplex algorithm with **Bland's pivot rule**
- Both terminate after a finite number of pivoting operations, since the number of bases is finite and Bland's pivot rule prevents cycling

Proof (2/4)

Point 1.

- If the LP problem admits a feasible solution, then, by the theorem on the feasibility of an LP problem, the optimal value of the simplex auxiliary problem is 0
- The Phase 0 terminates with an optimal dictionary of the simplex auxiliary problem in which, possibly after one further pivoting operation, the auxiliary variable x_0 is non-basic
- The associated partition $\mathcal{B}$ and $\mathcal{N}$ is then a basis of the LP problem whose basic solution

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \geq 0$$

is feasible

Proof (3/4)

Point 3. If the LP problem is not infeasible, by Point 1 the simplex algorithm can be started from a basic feasible solution. It terminates in one of two ways:

(i) with a dictionary in which $\bar{c}_j \leq 0$ for every $j \in \mathcal{N}$: by the sufficient conditions for optimality, the associated basic feasible solution is an optimal solution;

(ii) with a non-basic variable x_p such that $\bar{c}_p > 0$ and $\bar{a}_{ip} \leq 0$ for every $i \in \mathcal{B}$: by the certificate of unboundedness, the LP problem is unbounded.

Accordingly, the LP problem is infeasible, unbounded, or it admits an optimal solution, and the three cases are mutually exclusive.

Proof (4/4)

Point 2.

- If the LP problem admits an optimal solution, then it is neither infeasible nor unbounded
- Then the simplex algorithm terminates in case (i) of Point 3, with an optimal solution which is a basic feasible solution

□

An algorithmic consequence

- The Fundamental Theorem of Linear Programming has an important algorithmic consequence: to solve an LP problem it is enough to look for an optimal solution among its **basic** feasible solutions
- Their number is finite and at most

$$\binom{n+m}{m}$$

- The simplex algorithm explores a small subset of them, moving from one basic feasible solution to another one whose objective value is not smaller

Summary: the Fundamental Theorem of Linear Programming

The theorem:

- Feasible $\Rightarrow$ a basic feasible solution exists
- Optimal solution $\Rightarrow$ an optimal basic feasible solution exists
- Every LP problem is infeasible, unbounded, or has an optimal solution

The proof:

- Phase 0 and simplex algorithm with Bland's pivot rule terminate
- Their outcomes are exactly the three cases

Summary (1/2)

- When $b_i < 0$ for some i, the slack variables do not give a basic feasible solution and the **Phase** 0 is executed
- The **simplex auxiliary problem** adds $x_0 \geq 0$, subtracts it from every constraint and maximizes $-x_0$: it is never infeasible and never unbounded
- The LP problem is feasible if and only if the optimal value of the simplex auxiliary problem is 0
- A **special pivoting operation** (x_0 enters, the most negative basic variable leaves) makes the first dictionary of the simplex auxiliary problem feasible

Summary (2/2)

- At the end of the Phase 0: optimal value < 0 (infeasible), or 0 with x_0 non-basic, or 0 with x_0 basic (one more pivot)
- The initial dictionary of the LP problem is obtained by removing x_0 and writing the original objective function z in function of the non-basic variables
- The **Fundamental Theorem of Linear Programming**: every LP problem is infeasible, unbounded, or has an optimal solution, and then also an optimal **basic** feasible solution
- It is enough to search among the at most $\binom{n+m}{m}$ basic feasible solutions